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Case Study: Network Design for Venus Technologies

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1. Introduction

Venus Technologies is a new IT consulting company based in Windhoek, Namibia. As the company grows, it is opening five more branches in Oshakati, Grootfontein, Mariental, Lüderitz and Opuwo. We were hired as the network engineering team for this project. Our job was to design, address, configure and test a network in Cisco Packet Tracer that connects all six branches together, keeps every department on its own subnet, and can still work as the company grows from 505 employees today to a projected 2 020 employees by 2030.

In this document we explain, in our own words, every decision we made: why we designed the network the way we did, how we split up the IP address block we were given, how we configured every device, and how we proved that the finished network actually works. We have written this so that any one of us can use it to explain the project during the presentation, even the parts a teammate configured.

2. Design factors, goals and assumptions

2.1 Factors we considered

- How many employees each branch has today, and how those employees split across the three departments (Sales, Finance, IT) — this decided how big each department's subnet needed to be.
- Future growth: the case study tells us the company expects to grow from 505 to 2 020 employees by 2030 (roughly four times bigger). Because of this we did not size subnets exactly to today's headcount — we left headroom inside each subnet and reserved extra, completely separate subnets that are not used yet, so the network does not have to be redesigned from scratch as the company grows.
- Which branches were allowed to use VLANs. The brief only allows four branches to use VLANs, so we had to choose which four. We chose the four biggest branches by headcount — Windhoek, Oshakati, Lüderitz and Mariental — because they have the most employees and therefore benefit the most from the extra traffic separation and organisation that VLANs give. Opuwo and Grootfontein are the two smallest branches, so separating their three departments onto three physical router interfaces instead of VLANs is simple enough and still meets the requirement that each department has its own subnet.
- Security of the network devices themselves, so that a person who is not the network administrator cannot log in and change the configuration of a router or switch.
- Secure remote management, since the administrator is based in Windhoek and needs to be able to reach and configure every branch router without physically travelling to each town.
- Loop prevention and redundancy on the four switched branches, using the Spanning Tree Protocol (STP).

2.2 Goals

- Every department, in every branch, must be able to communicate with every other department in every other branch.
- The addressing plan must still work without being redesigned as the company grows to 2 020 employees.
- The network must be secure against unauthorised access, and must still allow the administrator to manage it remotely and safely.

2.3 Assumptions

- We assumed that showing communication from one representative PC per department is enough to prove the design works, instead of connecting all 505+ employees' computers individually in Packet Tracer.
- We assumed the company will eventually get a real public IP address from an actual Internet Service Provider for the Windhoek-to-ISP link. Since we do not have a real ISP, we used 203.0.113.0/30 to represent that link. This range is not a random guess — it is part of 203.0.113.0/24, a block that RFC 5737 sets aside specifically for use in documentation and examples, so it is the correct, standard placeholder to use instead of inventing a random public-looking address.
- We assumed each branch will keep growing at roughly the same steady rate rather than one branch suddenly exploding in size, so we reserved one extra subnet of the same size in every branch, rather than making some branches' reserved space bigger than others.

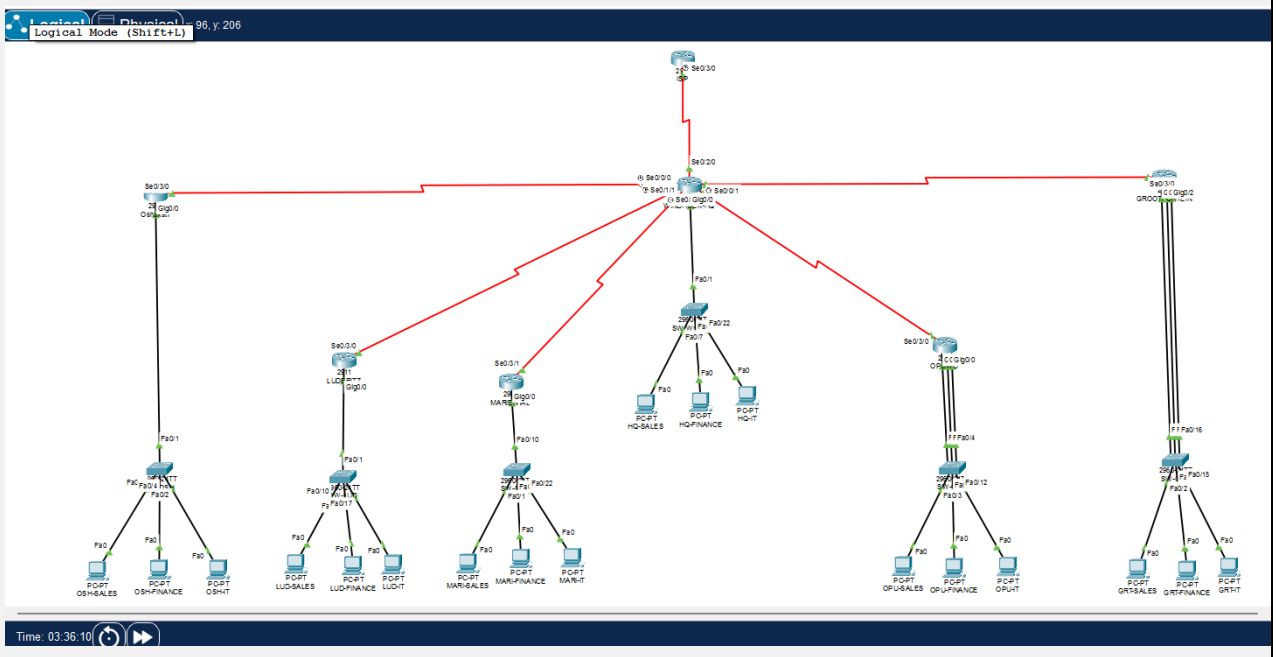
3. Network topology

We used a hub-and-spoke (star) topology. The Windhoek router (R-HQ) is the hub: it is the only router that connects to the Internet Service Provider, and it also has a direct, dedicated point-to-point link to each of the five branch routers. This means every branch only needs one WAN link back to HQ, and HQ is the single place where all inter-branch traffic passes through and gets routed.

Windhoek, Oshakati, Lüderitz and Mariental each use one switch to separate their Sales, Finance and IT departments into three VLANs. Opuwo and Grootfontein connect their three departments directly to three separate router interfaces instead, without VLANs, because the case study only requires four of the six branches to use VLANs — using VLANs on all six was not needed and would have added configuration we were not asked for.

Figure 1

This screenshot proves: The complete topology with every router, switch and PC placed, correctly named, correctly cabled, and every link showing a solid green light in Packet Tracer.



3.1 Devices used

We used two different router models on purpose, not the same model everywhere. R-HQ is a Cisco 2911, while every branch router is a smaller Cisco 1941. HQ needs far more physical ports than a branch does — it must connect to the ISP and to all five branches at once, which is six separate WAN connections, plus its own three departments. The 2911 has more expansion slots, so it can physically hold enough add-on modules for all of those connections. A branch router only ever needs two connections (one link up to HQ, one link down to its own local switch), so the smaller, cheaper 1941 is enough there. Using the bigger router only where the extra capacity is actually needed, instead of using the same expensive model everywhere, is how a real network engineer would design this, and it keeps the cost of the project down.

Branch	Router	Switch	PCs	VLANs used
Windhoek (HQ)	R-HQ (2911)	SW-HQ	PC-HQ-Sales, PC-HQ-Finance, PC-HQ-IT	10, 20, 30
Oshakati	R-OSH (1941)	SW-OSH	PC-OSH-Sales, PC-OSH-Finance, PC-OSH-IT	10, 20, 30
Lüderitz	R-LUD (1941)	SW-LUD	PC-LUD-Sales, PC-LUD-Finance, PC-LUD-IT	10, 20, 30
Mariental	R-MAR (1941)	SW-MAR	PC-MAR-Sales, PC-MAR-Finance, PC-MAR-IT	10, 20, 30
Opuwo	R-OPU (1941)	SW-OPU	PC-OPU-Sales, PC-OPU-Finance, PC-OPU-IT	none (direct router ports)
Grootfontein	R-GRT (1941)	SW-GRT	PC-GRT-Sales, PC-GRT-Finance, PC-GRT-IT	none (direct router ports)

We also had to add extra port modules to several routers before cabling, because the routers do not come with enough built-in ports for everything they connect to:

- R-HQ needed six serial-capable ports in total (one to the ISP and one to each of the five branches), so we powered it off and added HWIC-2T serial modules until it had enough slots. Without this, HQ physically could not have six separate WAN cables plugged into it at once.
- R-OSH, R-LUD and R-MAR each needed one extra serial port added for their single WAN link back to HQ, since their onboard port is already used for the trunk link down to their own switch.
- R-OPU and R-GRT each needed one extra serial port for their WAN link, and one extra LAN port. This is because these two branches do not use VLANs — each of their three departments needs its own separate physical router interface instead of sharing one interface through a trunk, so we needed a third LAN port on top of the two the router already has built in.

Whenever we added or removed a module on a router, we first switched the router off using the power button in the physical view, added the module, and then switched it back on — Packet Tracer (like a real router) will not let you add hardware modules while the device is powered on.

3.2 Cabling — what cable goes where, and why

- PC to switch: copper straight-through cable. This is the normal cable for connecting two different types of device (an end device and a switch), and it is what real network cabling standards call for in this situation.
- Switch to router: copper straight-through cable, for the same reason — the switch and the router are different types of device, so a straight-through cable is correct here too.
- Router to router, and the Windhoek-to-ISP link: serial DCE/DTE cable. We used serial links here (instead of copper Ethernet) because these are long-distance, point-to-point WAN links between sites, which is exactly what a serial connection represents in Packet Tracer — this mirrors how real branch offices are connected over a leased line or similar WAN service, not a direct Ethernet cable.

Every serial cable has a DCE end and a DTE end, and only the DCE end needs a clock rate configured on it (explained in section 5.2). We put the DCE end on R-HQ for all five branch links, and on the ISP router for the HQ-to-ISP link, and we checked this by clicking each cable once it was placed.

4. IP addressing scheme

We were given the address block 172.27.0.0/19. We divided it using VLSM (Variable Length Subnet Masking), which means every subnet is only as big as it actually needs to be, instead of giving every department the same fixed-size block regardless of how many people work there. This is more efficient and is the correct professional approach — it leaves far more address space free for the block's owner to use elsewhere or reserve for later.

We decided how big each department's subnet should be by looking at that branch's current headcount, splitting it evenly across the three departments, and then rounding up to a subnet size that gives comfortable room to grow without being wasteful:

- Windhoek has 255 employees today, about 85 per department. We gave Windhoek's three departments a /25 each (126 usable addresses), which is about 48% more than the current headcount per department.
- Oshakati (90 employees) and Lüderitz (72 employees) split to roughly 30 and 24 employees per department. We gave both of these branches /26 subnets (62 usable addresses each) — more than double their current per-department headcount.
- Mariental (33), Opuwo (29) and Grootfontein (27) split to roughly 11, 10 and 9 employees per department. We gave these three branches /27 subnets (30 usable addresses each), which is generous headroom without wasting large blocks of unused addresses on the smallest branches.

On top of this, every branch also has one extra reserved subnet, the same size as its department subnets, sitting immediately after its three department subnets and left completely unused for now. This is what satisfies the case study's requirement to "consider the growth of the company by reserving some subnets": if a branch adds a fourth department, or one department outgrows its current subnet, there is already space set aside for it without having to re-plan the whole address block. Beyond that, everything from 172.27.6.0 upward — well over half of the original /19 block — is left completely untouched, ready for entirely new branches the company may open after 2030.

Branch	Department	VLAN	Network (CIDR)	Usable range	Broadcast	Gateway
Windhoek (HQ)	Sales	10	172.27.0.0/25	.1 - .126	172.27.0.127	172.27.0.1
Windhoek (HQ)	Finance	20	172.27.0.128/25	.129 - .254	172.27.0.255	172.27.0.129
Windhoek (HQ)	IT	30	172.27.1.0/25	.1 - .126	172.27.1.127	172.27.1.1
Windhoek (HQ)	Reserved (growth)	—	172.27.1.128/25	—	172.27.1.255	—
Oshakati	Sales	10	172.27.2.0/26	.1 - .62	172.27.2.63	172.27.2.1
Oshakati	Finance	20	172.27.2.64/26	.65 - .126	172.27.2.127	172.27.2.65
Oshakati	IT	30	172.27.2.128/26	.129 - .190	172.27.2.191	172.27.2.129
Oshakati	Reserved (growth)	—	172.27.2.192/26	—	172.27.2.255	—
Lüderitz	Sales	10	172.27.3.0/26	.1 - .62	172.27.3.63	172.27.3.1
Lüderitz	Finance	20	172.27.3.64/26	.65 - .126	172.27.3.127	172.27.3.65
Lüderitz	IT	30	172.27.3.128/26	.129 - .190	172.27.3.191	172.27.3.129
Lüderitz	Reserved (growth)	—	172.27.3.192/26	—	172.27.3.255	—
Mariental	Sales	10	172.27.4.0/27	.1 - .30	172.27.4.31	172.27.4.1
Mariental	Finance	20	172.27.4.32/27	.33 - .62	172.27.4.63	172.27.4.33
Mariental	IT	30	172.27.4.64/27	.65 - .94	172.27.4.95	172.27.4.65
Mariental	Reserved (growth)	—	172.27.4.96/27	—	172.27.4.127	—
Opuwo	Sales	none	172.27.4.128/27	.129 - .158	172.27.4.159	172.27.4.129
Opuwo	Finance	none	172.27.4.160/27	.161 - .190	172.27.4.191	172.27.4.161
Opuwo	IT	none	172.27.4.192/27	.193 - .222	172.27.4.223	172.27.4.193
Opuwo	Reserved (growth)	—	172.27.4.224/27	—	172.27.4.255	—
Grootfontein	Sales	none	172.27.5.0/27	.1 - .30	172.27.5.31	172.27.5.1
Grootfontein	Finance	none	172.27.5.32/27	.33 - .62	172.27.5.63	172.27.5.33
Grootfontein	IT	none	172.27.5.64/27	.65 - .94	172.27.5.95	172.27.5.65
Grootfontein	Reserved (growth)	—	172.27.5.96/27	—	172.27.5.127	—

4.1 WAN links

Every WAN link only ever has exactly two devices on it — the router at each end — so we used a /30 subnet for each one. A /30 gives exactly two usable host addresses, which is the smallest subnet size that can do the job, so it wastes the least amount of address space on links that will never need more than two addresses.

Link	Network	HQ / ISP side IP	Branch side IP
R-HQ – R-OSH	172.27.5.128/30	172.27.5.129	172.27.5.130
R-HQ – R-LUD	172.27.5.132/30	172.27.5.133	172.27.5.134
R-HQ – R-MAR	172.27.5.136/30	172.27.5.137	172.27.5.138
R-HQ – R-OPU	172.27.5.140/30	172.27.5.141	172.27.5.142
R-HQ – R-GRT	172.27.5.144/30	172.27.5.145	172.27.5.146
R-HQ – ISP	203.0.113.0/30 (ISP-assigned)	203.0.113.1	203.0.113.2 (ISP)

We deliberately used a separate documentation-range address (203.0.113.0/30) for the HQ-to-ISP link instead of another 172.27.x.x subnet, because in real life the company would not own the public IP address on the ISP side of that link — the ISP would assign it. Keeping it outside our own 172.27.0.0/19 block makes that distinction clear in the documentation.

5. Device configuration

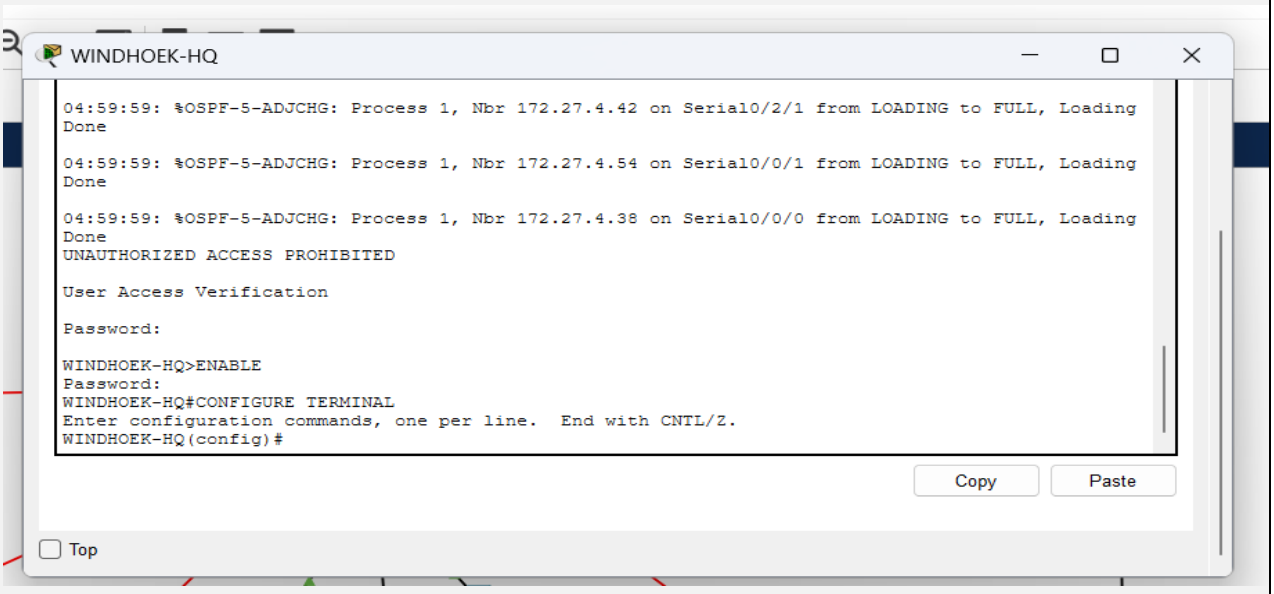
5.1 Device access security

We configured the same security settings on every single router and switch (13 devices in total), changing only the hostname each time, so that every device is equally protected:

- An enable secret password, which protects privileged (admin) mode — without it, anyone could type "enable" and get full control of the device.
- A console password, which protects anyone connecting through the physical console port.
- A password on the VTY (virtual terminal) lines, which protects remote logins.
- service password-encryption, which scrambles all the passwords stored in the configuration file so that someone who views the config cannot simply read the passwords in plain text.
- exec-timeout 5 0, set on the console and VTY lines. This automatically logs an idle session out after 5 minutes of no activity. We added this so that if an administrator walks away from a terminal without logging out, the session does not stay open and vulnerable indefinitely — it closes itself.
- A login banner (banner motd), which displays a warning message before login, telling anyone connecting that only authorised users are allowed. This is standard practice on real company networks, partly as a deterrent and partly because it can be used as legal proof that unauthorised access was clearly against policy.

Figure 2

This screenshot proves: The CLI of one device (R-HQ) showing the login banner appearing, and the enable secret successfully asking for a password before granting privileged mode. This proves the device access security requirement is actually working.



5.2 Router interface configuration

We configured every router interface using the IP addresses from the addressing tables in section 4. On the WAN links, only the DCE end needs a clock rate command — this is because in a real serial connection, one side (the DCE, usually the service provider's equipment) is responsible for generating the timing signal that both ends use to synchronise their data, while the other side (the DTE) just uses whatever clock rate the DCE provides. Without a clock rate set on the DCE end, the serial link will not come up.

On the four VLAN branches (Windhoek, Oshakati, Lüderitz, Mariental) we used router-on-a-stick: one single physical interface on the router was split into three logical sub-interfaces (for example Gi0/0.10, Gi0/0.20, Gi0/0.30), each carrying one VLAN's traffic and each configured with the IP address that acts as the default gateway for that department. We chose this instead of giving each department its own physical cable back to the router because it needs far fewer physical ports and cables while still keeping every department's traffic on its own separate subnet.

On Opuwo and Grootfontein, since these two branches do not use VLANs, each department connects to its own separate physical router interface instead (this is why these two routers needed an extra LAN port added, as explained in section 3.1). Each physical interface is simply given the IP address of that department's gateway directly, with no sub-interfaces or trunking involved.

Figure 3a

This screenshot proves: "show ip interface brief" on R-HQ, with every interface showing status "up" and protocol "up", and the IP addresses matching the addressing tables above.

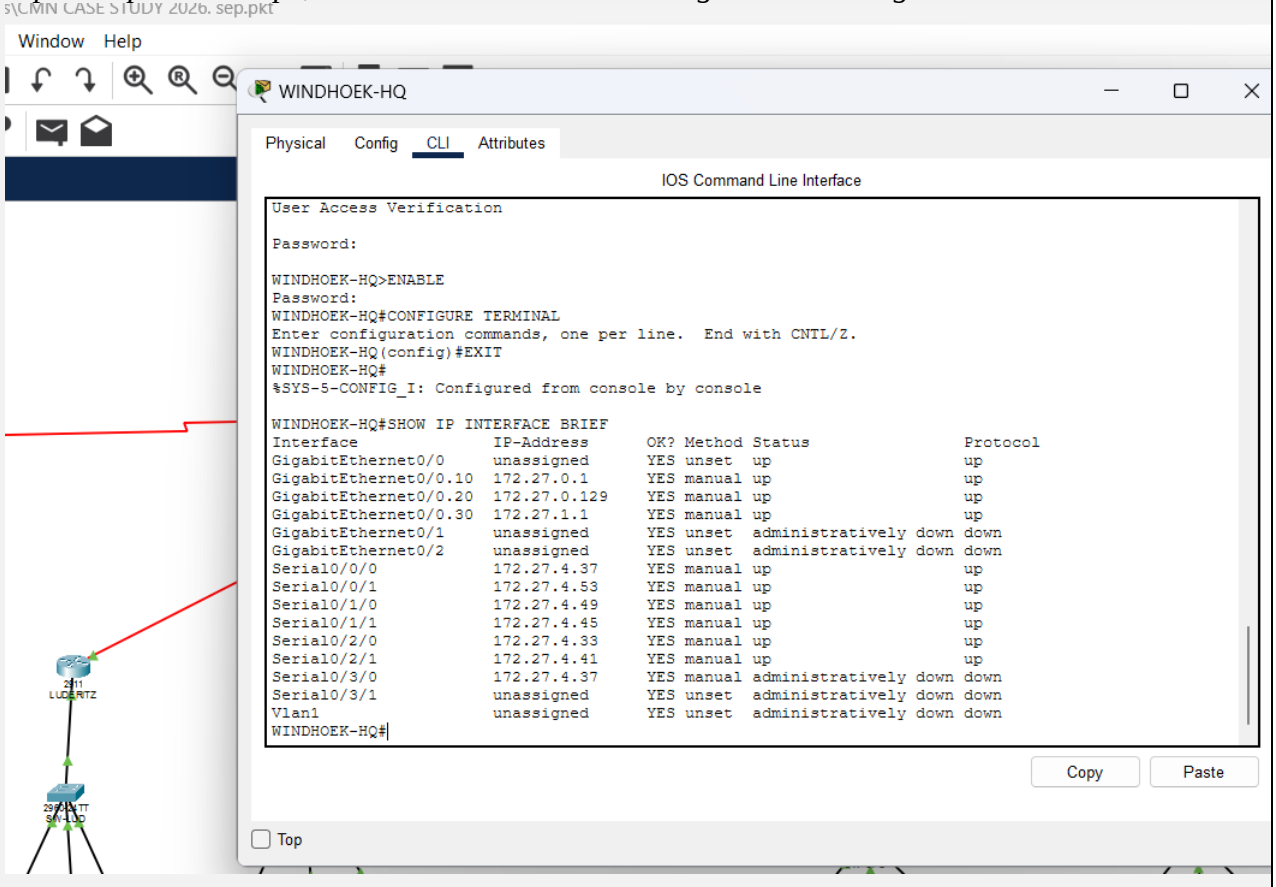


Figure 3b

This screenshot proves: "show ip interface brief" on R-OSH, confirming its WAN link to HQ and its three VLAN sub-interfaces are all up and correctly addressed.

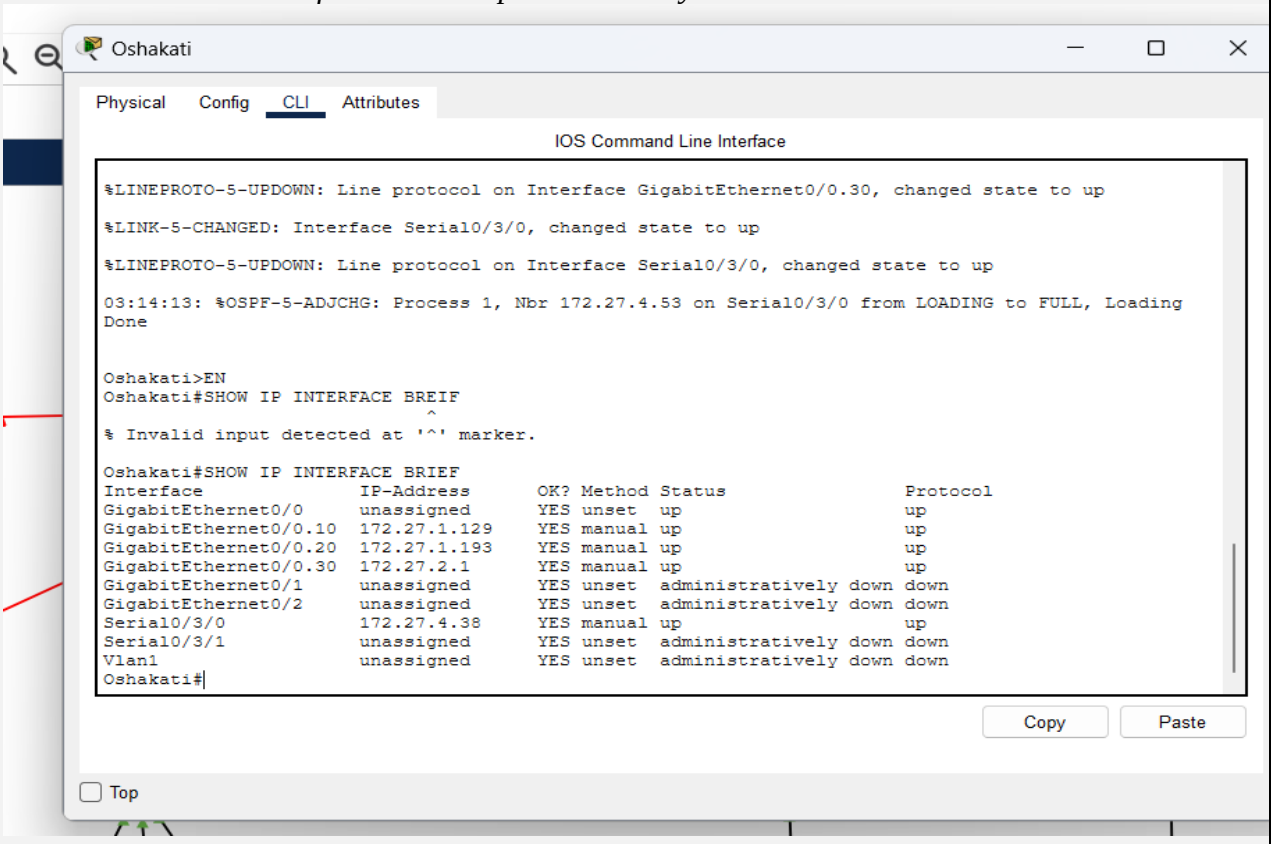


Figure 3c

This screenshot proves: "show ip interface brief" on R-LUD, confirming the same for Lüderitz.

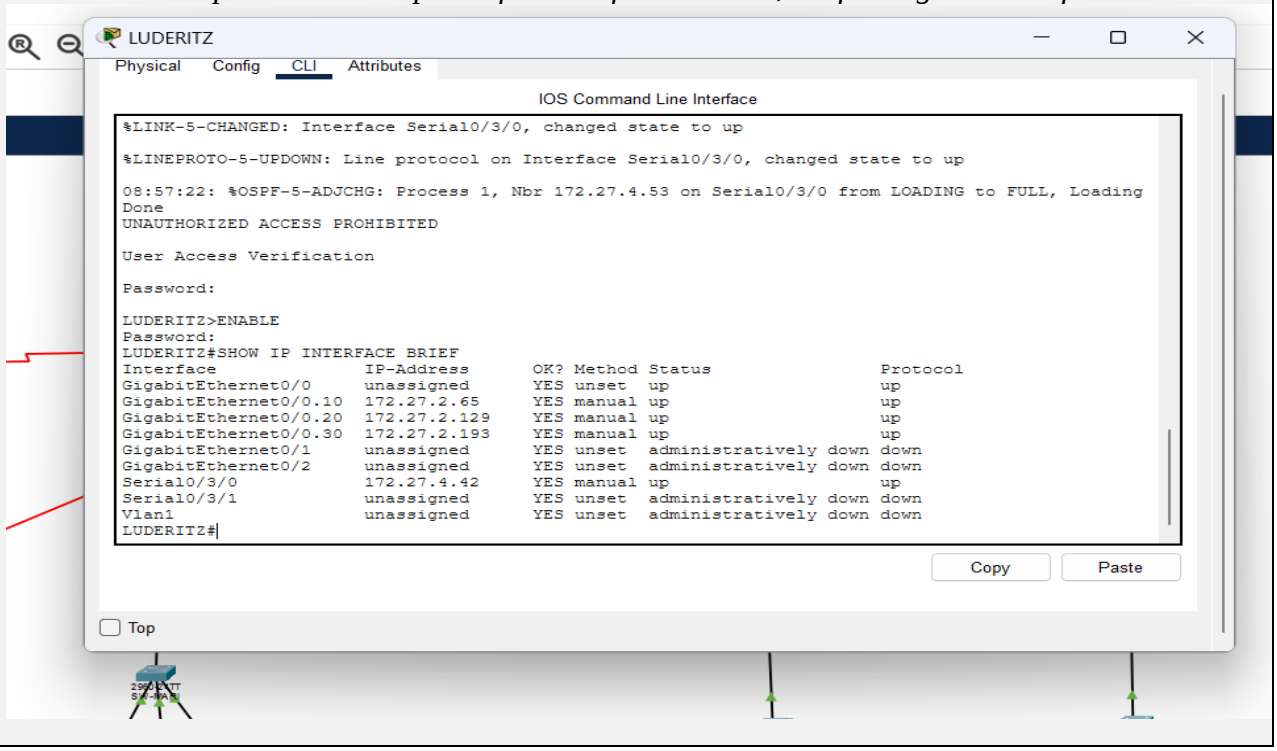


Figure 3d

This screenshot proves: "show ip interface brief" on R-MAR, confirming the same for Mariental.

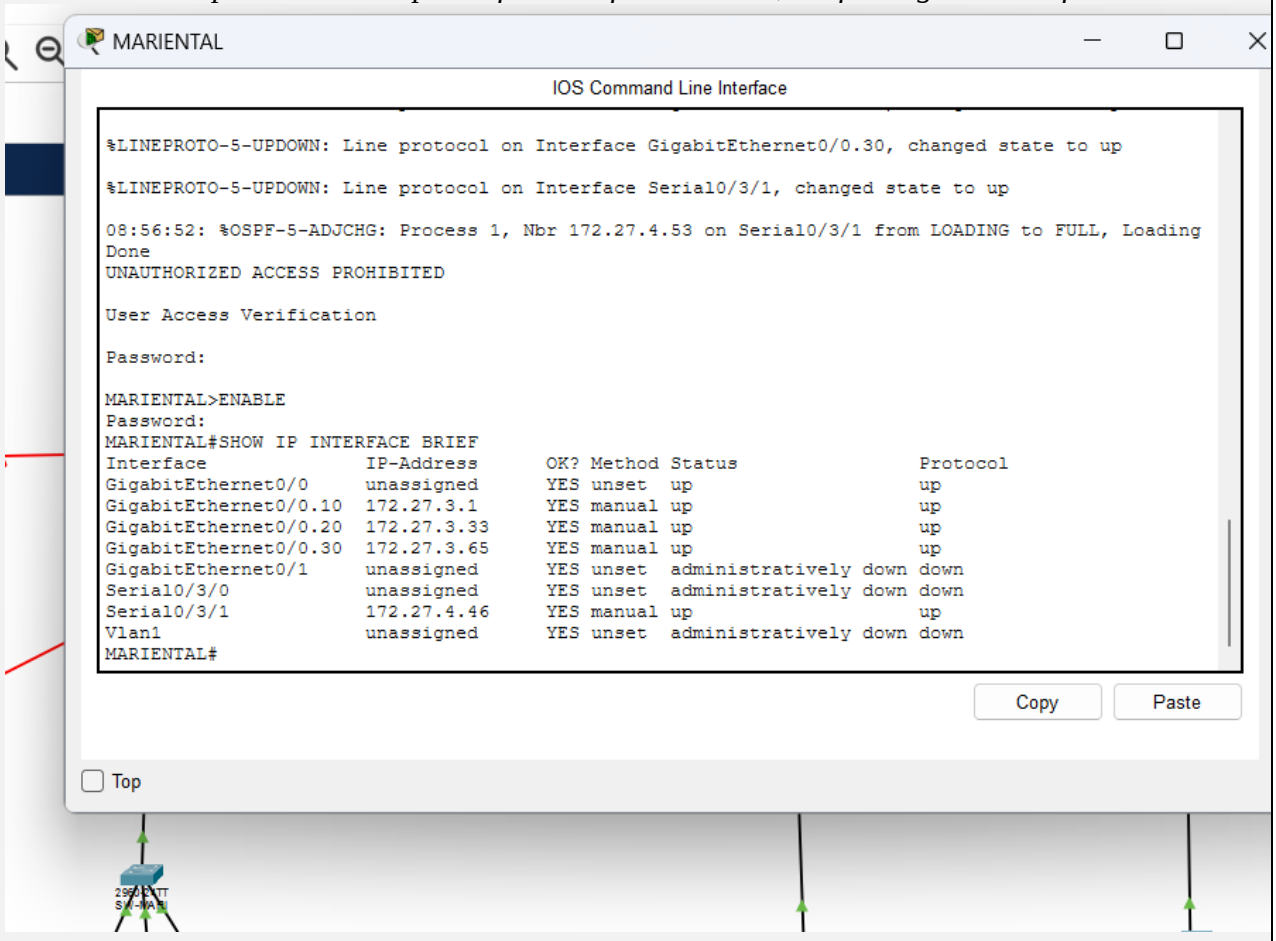


Figure 3e

This screenshot proves: "show ip interface brief" on R-OPU, confirming its WAN link and its three separate physical department interfaces are all up and correctly addressed.

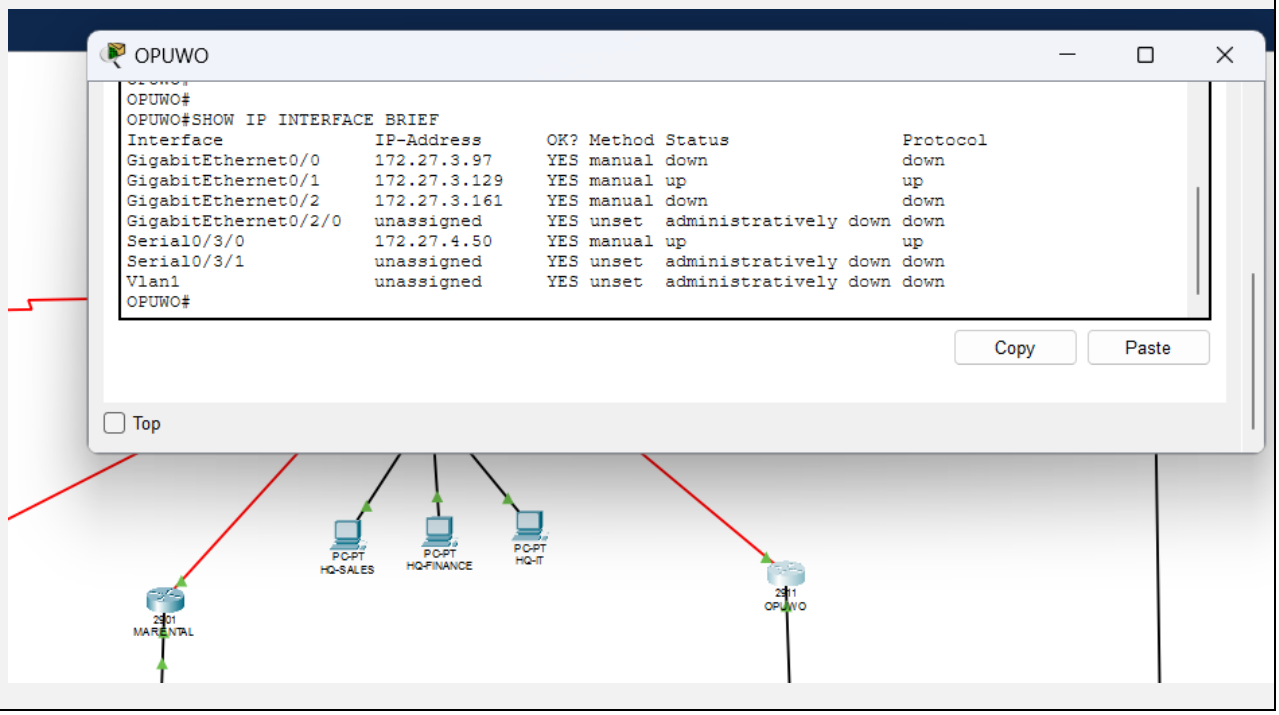
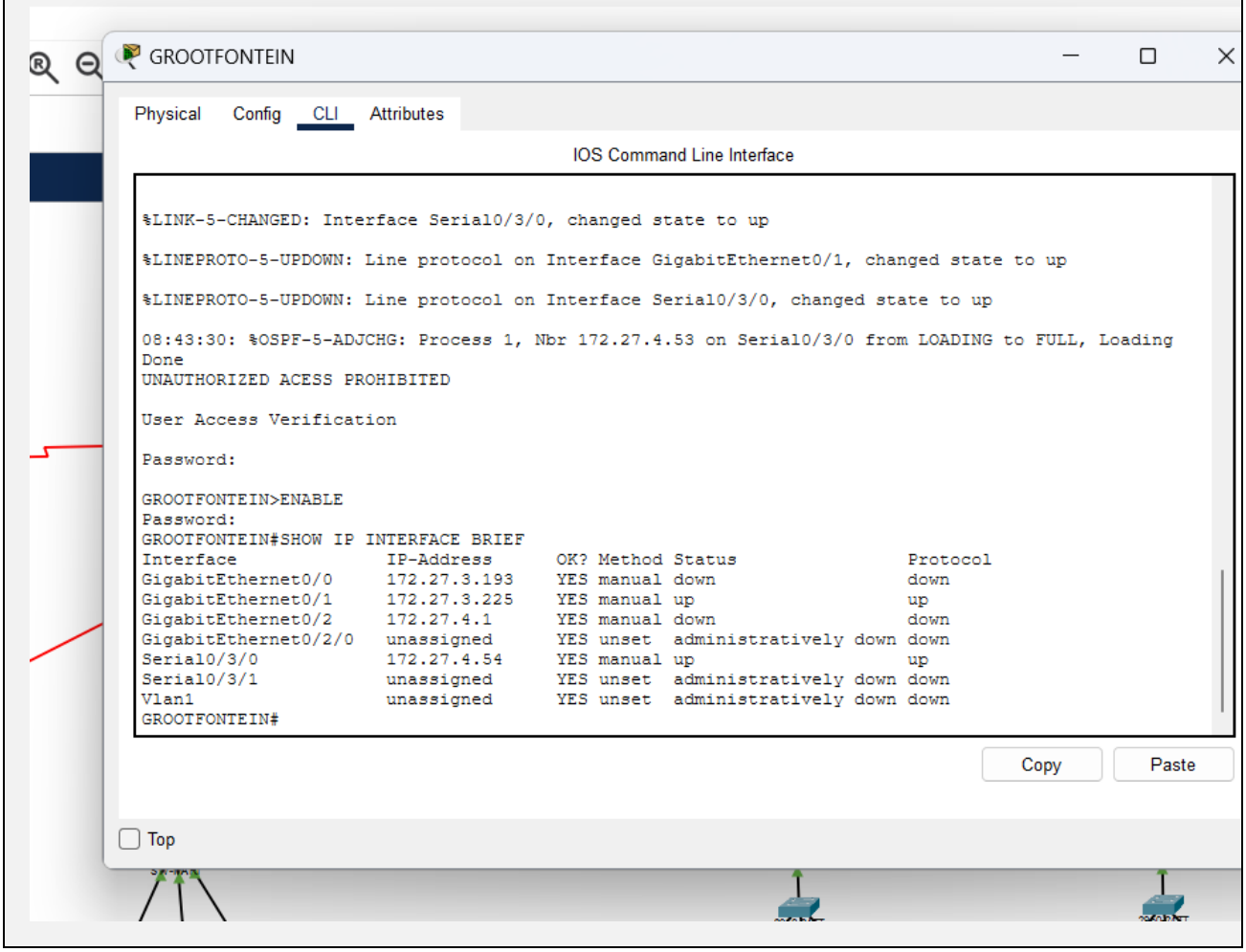


Figure 3f

This screenshot proves: "show ip interface brief" on R-GRT, confirming the same for Grootfontein.



6. VLAN configuration and security on switches

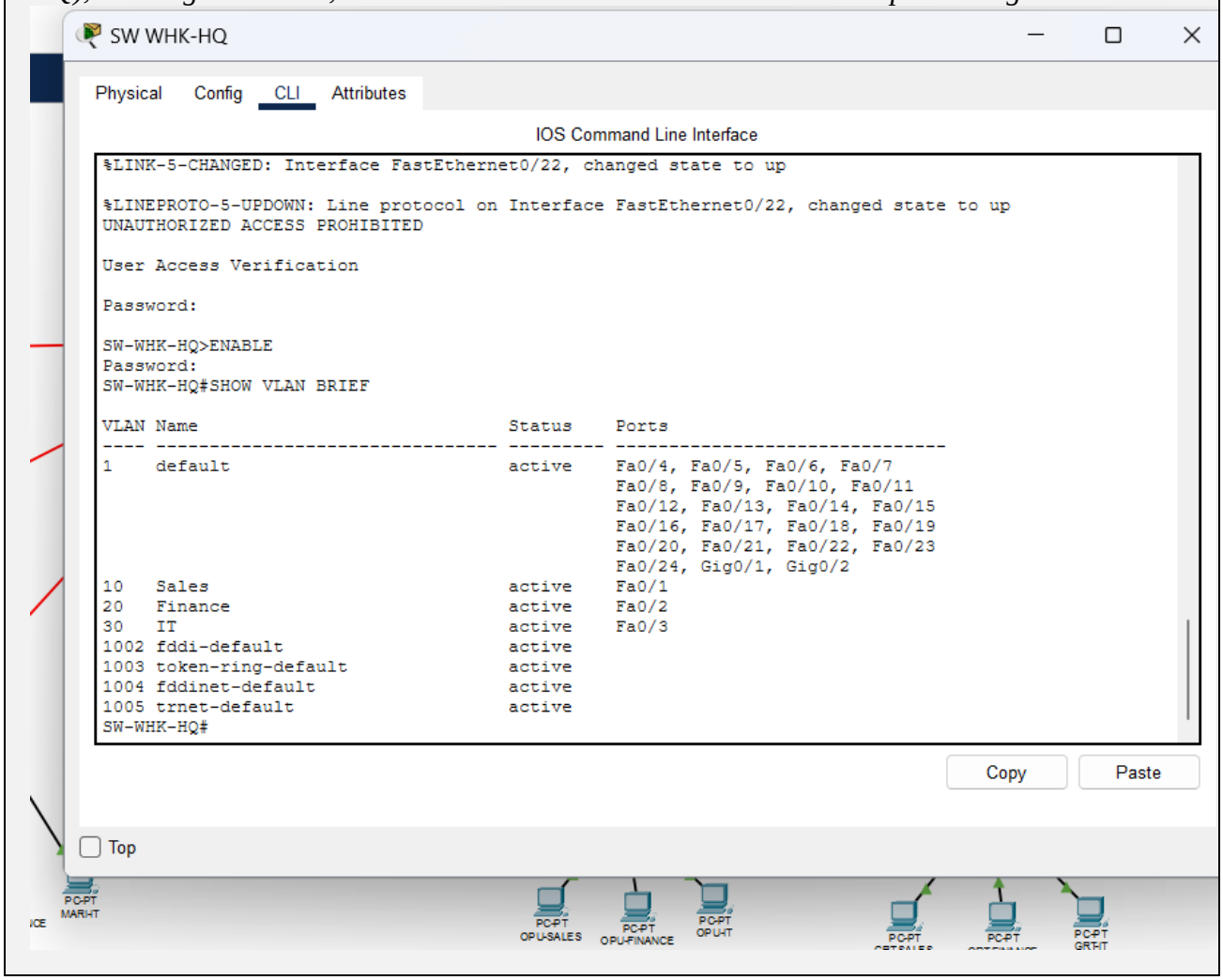
On SW-HQ, SW-OSH, SW-LUD and SW-MAR we created three VLANs: VLAN 10 for Sales, VLAN 20 for Finance and VLAN 30 for IT. We used the same VLAN numbers on every one of the four branches so that the numbering is consistent and easy to remember company-wide — VLAN 10 always means Sales, no matter which branch you are looking at. We numbered them in tens (10, 20, 30) instead of 1, 2, 3, so that there is room to insert new VLANs later (for example VLAN 11 for a second Sales team) without having to renumber the existing ones.

We assigned each switch access port to the correct VLAN depending on which department's PC is plugged into it, and we configured the single port going up to the router as a trunk port carrying only VLANs 10, 20 and 30. This is also where the "security" part of this section comes in: by setting access ports to a single, specific VLAN each, a device plugged into the Sales port cannot see Finance or IT traffic even though it is on the same physical switch. And by explicitly restricting the trunk to only the three VLANs we actually use (switchport trunk allowed vlan 10,20,30), we stop any other, unused VLAN from ever being able to cross that trunk link, which closes off a common way attackers try to "VLAN hop" between departments that are supposed to be kept separate.

On Opuwo and Grootfontein we did not create any VLANs at all, since these two branches separate their departments using direct physical router interfaces instead, as explained in section 5.2.

Figure 4

This screenshot proves: "show vlan brief" on one of the four VLAN switches (for example SW-HQ), showing VLANs 10, 20 and 30 each listed with the correct switch ports assigned to them.



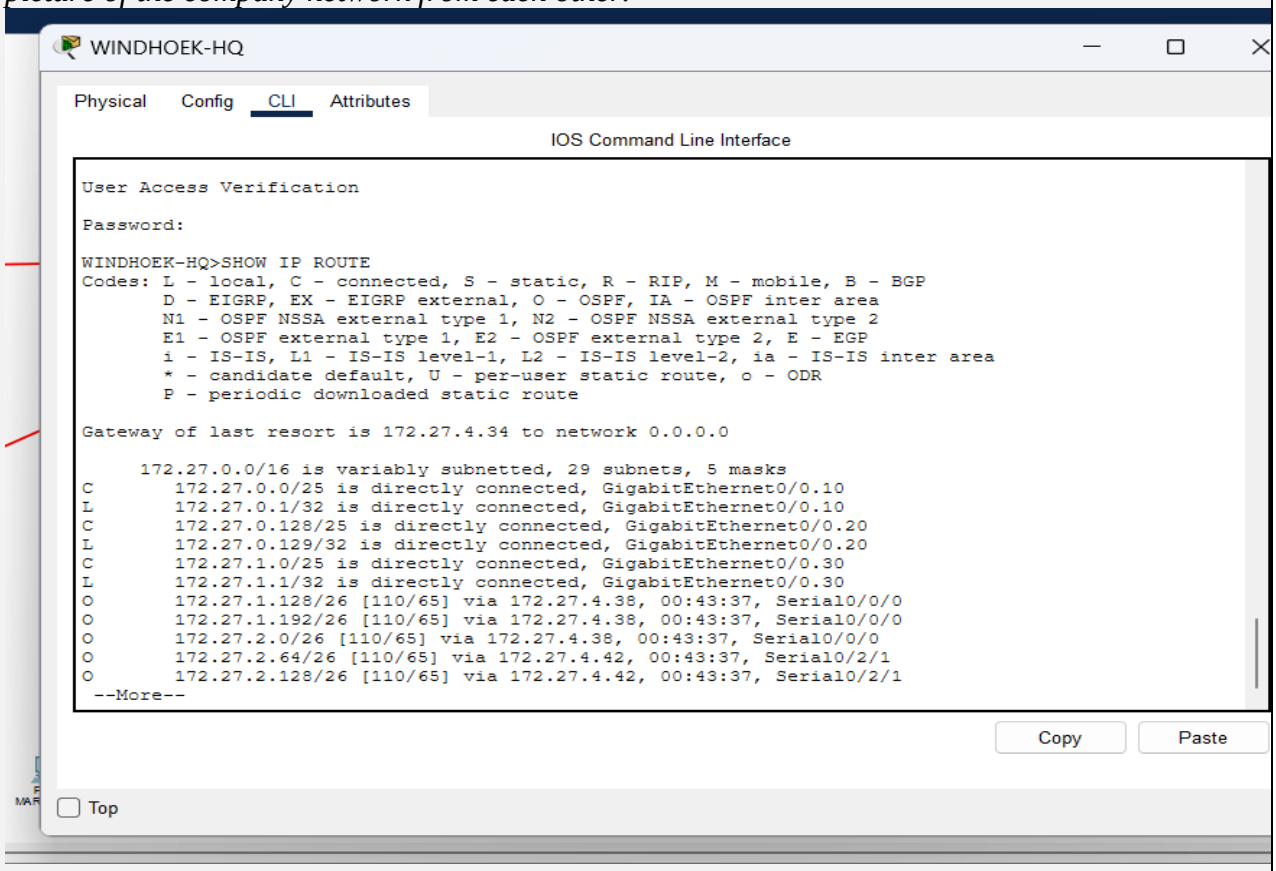
7. Routing protocol configuration

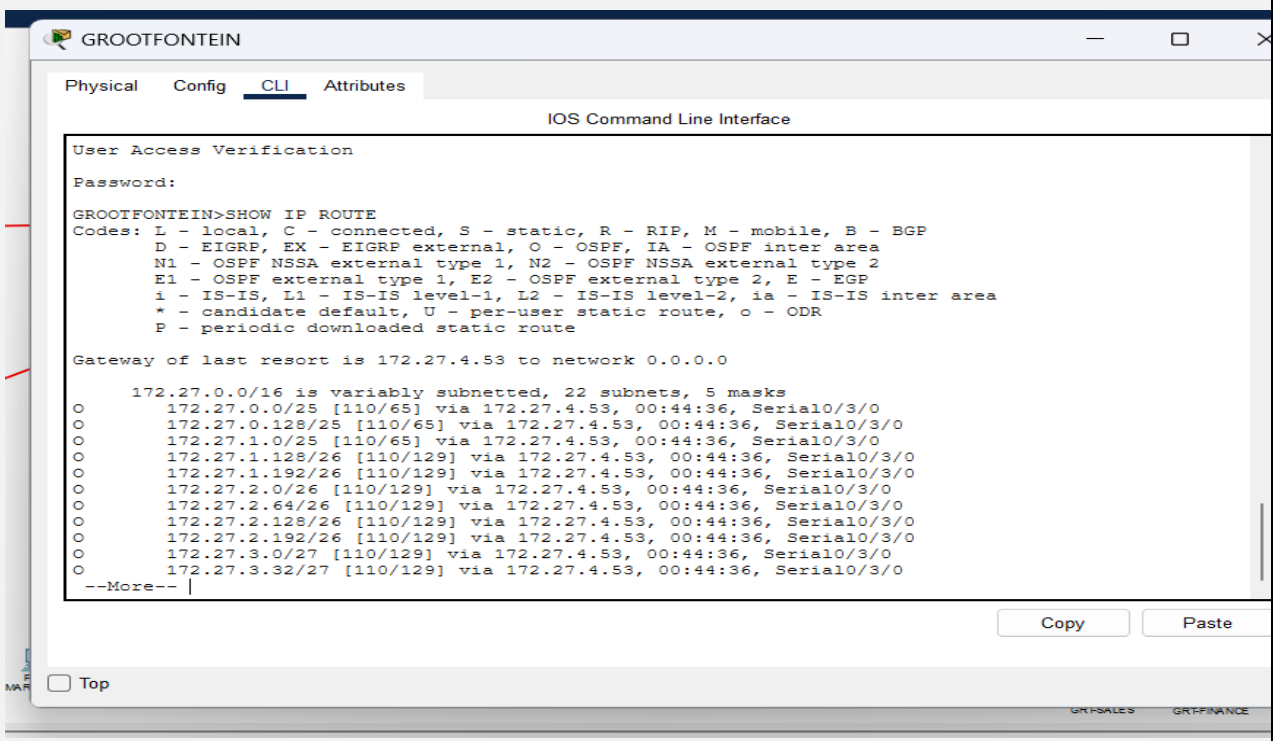
We chose OSPF (Open Shortest Path First), running in a single area (area 0), as our routing protocol. We chose OSPF for a few reasons: our topology is a simple hub-and-spoke design, which OSPF handles very well; OSPF converges quickly if a link goes down, meaning the network finds a new path fast instead of staying broken; and it is a standard, widely used, open (non-proprietary) protocol, which means any engineer taking over this network later would recognise it immediately. We kept everything inside one single area because our network is small — multi-area OSPF exists to keep very large networks organised and is not necessary here, so adding it would only have made the configuration more complicated without any real benefit.

We configured OSPF on all six routers so that each one advertises every network it is directly connected to. On R-HQ specifically, we also added a default route pointing out to the ISP, and told OSPF to advertise that default route to the rest of the network (default-information originate). This means that if any department ever needed to reach something outside our own network (for example the actual internet), every router already knows to send that traffic to R-HQ, which is the only router actually connected to the ISP.

Figure 5

This screenshots proves: "show ip route" on two different routers, showing routes marked with "O" (meaning learned through OSPF) to the other branches' subnets — this proves the routers are not just directly connected to their own local networks, but have actually learned the full picture of the company network from each other.





8. Spanning Tree Protocol discussion

In the four branches that use switches (Windhoek, Oshakati, Lüderitz and Mariental), the Spanning Tree Protocol (STP) runs automatically in the background to protect the network from loops. If two switches in the same branch were ever connected to each other more than once — for example as a backup cable in case the main one fails — this would create a loop. A loop is a serious problem because broadcast frames would circulate around it forever, multiplying every time, which quickly floods the whole switched network and can bring it down completely (this is sometimes called a broadcast storm).

STP solves this by automatically detecting when more than one path exists between two switches and putting the extra link into a blocking state, so that only one active path is ever actually forwarding traffic at a time — this removes the loop. Importantly, STP does not disconnect the extra link, it just keeps it on standby. If the currently active link ever fails (a cable is unplugged, a switch port dies, and so on), STP notices this and automatically moves the blocked link into a forwarding state instead, so the network keeps working with little to no manual effort from the administrator. This is what gives our switched branches both loop prevention during normal operation, and redundancy if something goes wrong.

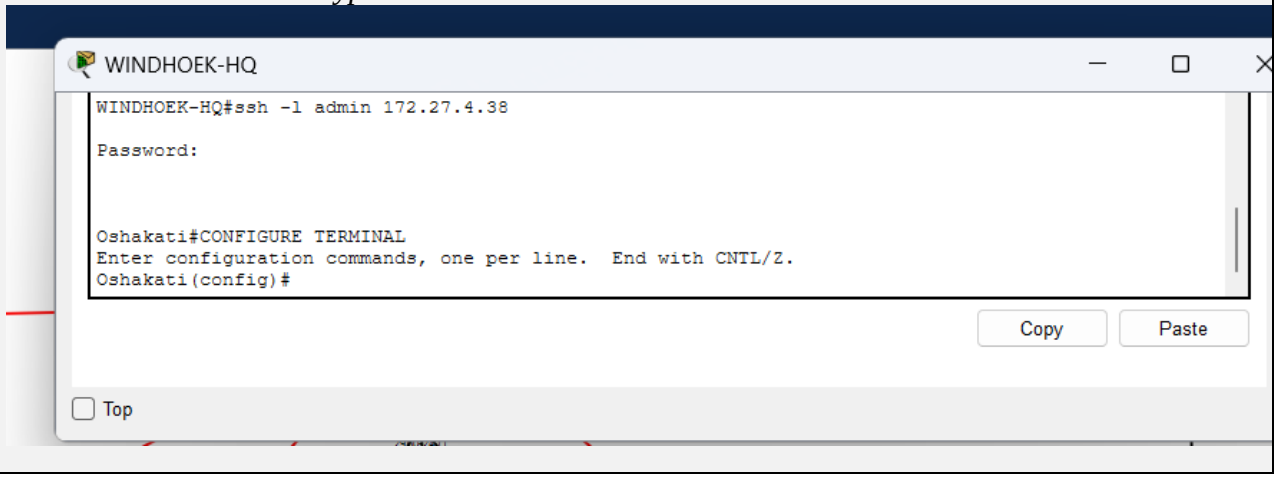
9. Secure remote management

The network administrator is based in Windhoek, so every router needs to be reachable and configurable from R-HQ without anyone having to physically travel to each branch every time a change is needed. We configured SSH (Secure Shell) instead of Telnet on all six routers, because SSH encrypts the entire session, while Telnet sends everything — including usernames and passwords — as plain, readable text. Anyone able to see the traffic on a Telnet session could simply read the administrator's password; with SSH this is not possible.

To enable SSH we set a domain name on each router (SSH needs this to generate its encryption keys), then generated an RSA key pair using a 1024-bit key length, which is the minimum key size Cisco IOS requires before it will allow SSH to be enabled. We then created a local username and password on each router, and set the VTY lines to accept only SSH connections (transport input ssh) authenticated against that local username database (login local), rather than continuing to accept plain Telnet or a simple line password. Finally, we tested this by connecting from R-HQ to a branch router using the ssh command and logging in successfully.

Figure 6

This screenshot proves: A successful SSH login from R-HQ into a branch router, showing the SSH prompt and the branch router's own hostname prompt appearing once logged in — this proves the administrator really can manage a remote router securely from Windhoek, not just that the commands were typed in. command: ssh –l admin 172.27.4.38

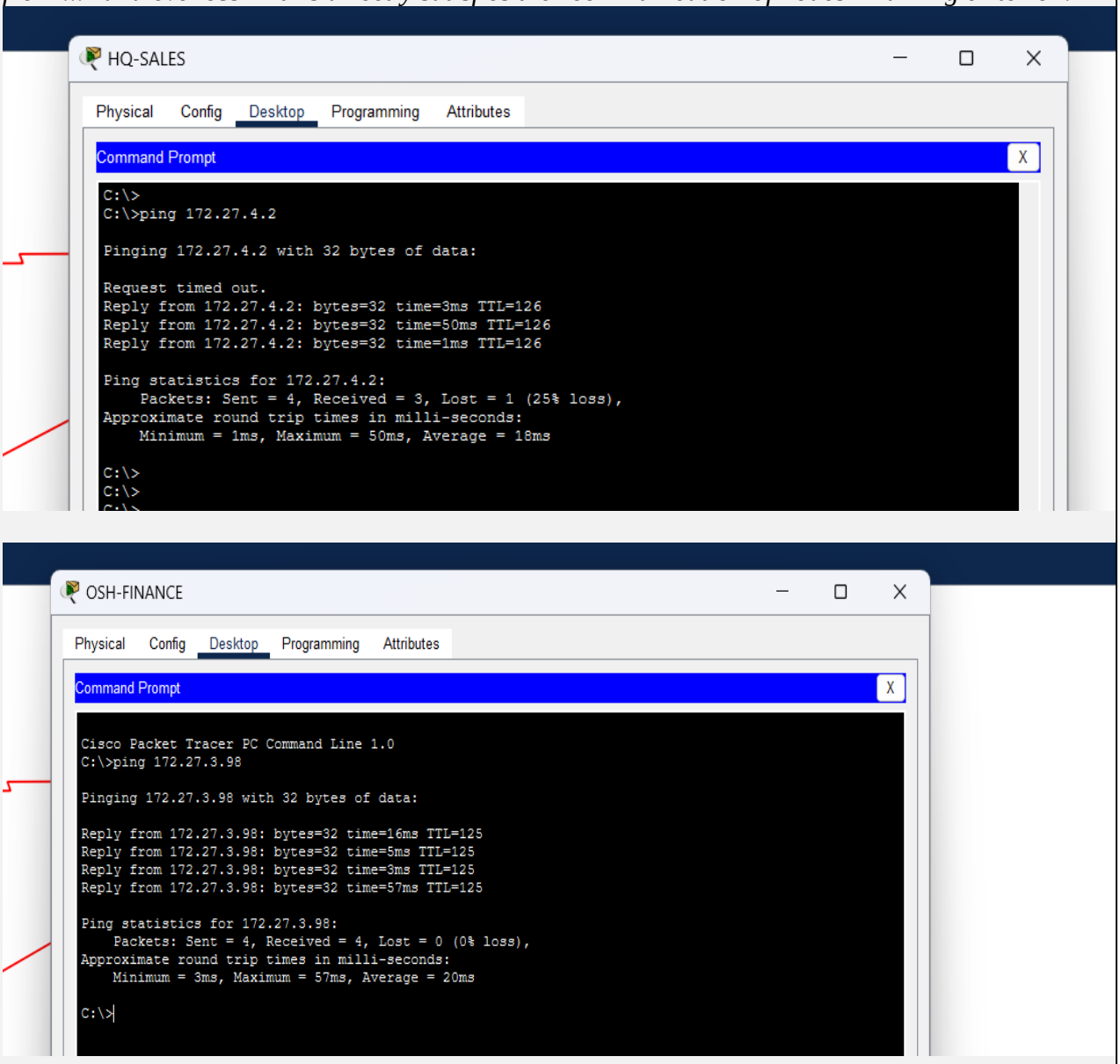


10. Communication testing

To prove that the whole network actually works end to end, and not just on paper, we used the ping command from PCs in different branches to reach PCs in other branches — deliberately choosing pairs of PCs whose traffic has to pass through several routers, and mixing VLAN branches with non-VLAN branches so that both parts of our design are tested, not just one. Every ping below completed successfully with 0% packet loss, which shows that our IP addressing, VLAN configuration, routing (OSPF) and cabling are all working together correctly at the same time — if any one of those had been set up wrong, these pings would have failed or timed out.

Figure 7

This screenshot proves: Two successful pings between PCs in different branches (for example PC-HQ-Sales to PC-GRT-IT, and PC-OSH-Finance to PC-OPU-Sales), each showing "Reply from ..." and 0% loss — this directly satisfies the "communication of nodes" marking criterion.



11. Conclusion

We successfully designed and built a network for Venus Technologies that connects all six branches, keeps every department on its own subnet, and leaves enough address space — both inside each branch and across the whole 172.27.0.0/19 block — for the company to grow toward 2 020 employees by 2030. We used VLANs on the four largest branches to keep department traffic organised and separated, and direct router interfaces on the two smallest branches, exactly as the brief required. We used OSPF so that every router learns the full network automatically and can reroute quickly if a link fails, and SSH so that the administrator in Windhoek can manage every router safely without travelling. We secured every device with passwords, encrypted the stored passwords, added an idle-session timeout and a login banner, and we explained how STP protects the four switched branches from loops while still giving them redundancy. Finally, we tested the finished network with ping across multiple branches and confirmed that every part of it can reach every other part.

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